

Francesco Cancelliere

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Education

PhD - AI Robot Navigation in Unstructured Environments Aug 2023 - July 2026
University of Catania

- Leading applied research focused on developing Deep Reinforcement Learning algorithms for outdoor ground vehicle autonomy, focusing on end-to-end LiDAR and RGB camera AI models.
- Engineered and deployed a multi-agent full training pipeline (Python/PyTorch) within an Unreal Engine-based simulator, achieving a 10x increase in training efficiency for LiDAR tasks and 7x for RGB camera tasks compared to the previous single-agent architecture.
- Validated policies through extensive hardware testing on off-road platforms, focusing on sim-to-real transfer and real-world system reliability.

MSc - Automation Engineering and Control of Complex System Oct 2019 - Oct 2021

University of Catania

Graduated with 110/110 with Honors.

BSc - Electronics Engineering Oct 2016 - Feb 2020
University of Catania

Work Experience

Robotics Research Engineer May 2025 - Present
RobotiXX - George Mason University

- Designed and built improved version of an RC-scale robot for autonomous outdoor navigation. Reduced the center of gravity by 20%, improving off-road capabilities and stability. Refactored the software stack from ROS to ROS 2, ensuring compatibility with the latest navigation software stack.
- Developed C++ embedded software for the an off-road differential-drive platform featuring an offset turret, integrating 3-axis stabilization (IMU-based) and real-time control logic via micro-ROS on a Teensy 4.0 MCU.

Edge AI Application Engineer Jun 2024 - Present
Deepsensing S.r.l. - Part Time

- Supervise vision models development, ensuring compatibility with edge platforms, such as NVIDIA Jetson, Hailo-8, Intel ARRIA 10.
- Conduct performance benchmarking and testing to evaluate models and determine efficiency of each hardware platform for the task.

Embedded Engineer Feb 2021 - Dec 2023
Faradex S.r.l.

- Provided tailored consulting and hands-on development for early-stage product development, specializing in embedded platforms.
- Accelerated time-to-market for 5+ products by designing and deploying low-level firmware and RTOS (FreeRTOS) solutions on resource-constrained platforms (STM32,ESP32), focusing on ultra-low power applications (sub mAh consumption).

Robotic Engineer Consultant May 2022 - Nov 2022
Rosys Group - University of Catania

- Led the development of the autonomous beach cleanup system (SandRo), delivering a fully integrated, full-stack robotic solution from proof-of-concept to field testing in 6 months.
- Engineered the high-level perception and manipulation stack using *ROS/Python*, integrating a *YOLOv5* object detection pipeline with a *UR10e* robotic arm via *MoveIt*.

Technical Skills

- Robotics & Autonomy: ROS 1/2, micro-ROS, Nav2, ros2_control, MoveIt, Gazebo, Unreal Engine, Docker
- Programming: *C++*, *Python* (OOP, NumPy, SciPy), *C* (Embedded), MATLAB, Linux Shell Scripting
- AI/ML & Edge Deployment: *PyTorch*, TensorFlow, Computer Vision (YOLO), Edge Optimization (*NVIDIA* Jetson, *Hailo* – 8 SDK, TensorRT), Model Quantization
- Embedded Systems: FreeRTOS, Microcontrollers (*STM32*, *ESP32*, *AVR*), Device Drivers (I²C, SPI, UART, CAN), Debugging (*JTAG*), Soldering
- Design & Tools: Git/GitHub, Linux (Advanced CLI), Autodesk Fusion 360 (CAD), Altium Designer (Schematics), Oscilloscope, Logic Analyzer

Personal Details

Languages Italian (Native), English (Fluent)
Certifications UAS Pilot Open A1/A3
Citizenship American, Italian

Selected Publications

1. Cancelliere, F. et al. (2024). SandRo, a Robot for Beach Waste Cleanup. In *European Robotics Forum*, pp. 269–273, Springer.
2. Vecchio, G. et al. (2024). MIDGARD: A Robot Navigation Simulator for Outdoor Unstructured Environments. In *European Robotics Forum*, pp. 120–125, Springer.
3. Faro, R. et al. (2025). Timely Predicting Power Mosfet Failure in EV: AI Based Metodology and Deployment at the Edge. In *2025 IEEE Symposia on Computational Intelligence for Energy, Transport and Environmental Sustainability (CIETES Companion)*, pp. 1–5, IEEE.